# N OSCILADORES ACOPLADOS

$q_k \quad k = 1, \dots, n$ coordenadas xeneralizadas

$\left.\frac{\partial V}{\partial q_k}\right|_{eq} = 0$ configuración de equilibrio

Enerxía Potencial $V = \frac{1}{2} \sum_{j,k}^{n} A_{jk}\, q_j\, q_k$ oscilacións harmónicas ao redor da posición de equilibrio

$\left.\frac{\partial^2 V}{\partial q_j \partial q_k}\right|_{eq}$ → constantes do resorte.

Enerxía cinética $T = \frac{1}{2} \sum_{j,k}^{n} m_{jk}\, \dot{q}_j\, \dot{q}_k$ Ligaduras fixas

Lagranxiana $L = \frac{1}{2} \sum_{j,k}^{n} \left( m_{jk}\, \dot{q}_j\, \dot{q}_k - A_{jk}\, q_j\, q_k \right)$

Ecuación do movemento $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k} \rightarrow \sum_{j=1}^{n} \left( m_{jk}\, \ddot{q}_j + A_{jk}\, q_j \right) = 0 \quad (k = 1, \dots, n)$

Ecuación característica $q_j = a_j e^{i\lambda t} \rightarrow \sum_{j=1}^{n} \left( A_{jk} - \lambda^2 m_{jk} \right) a_j = 0 \quad (k = 1, \dots, n)$

$\downarrow$

$(A - \lambda^2 m)\, \vec{a} = 0$

$A$, $m$: Matrices; $\vec{a}$: vector de ctes de integración $a_j$

Ecuación Determinante secular $a_j \neq 0 \rightarrow \det(A - \lambda^2 m) = 0$

determinante $n \times n$ con $n$ raíces $\lambda^2 = \omega_r^2 \quad r = 1, \dots, n$

$\lambda = \pm \omega_r$

## Solución:

$$q_j(t) = \sum_{r=1}^{n} \left( a_{jr}^{+} e^{i\omega_r t} + a_{jr}^{-} e^{-i\omega_r t} \right) \qquad j = 1, \dots, n$$

ou ben mediante o cambio $a_{jr}^{+} = a_{jr} e^{-i\delta_r}$

$$a_{jr}^{-} = a_{jr}^{+*}$$

$$\boxed{q_j(t) = \sum_{r=1}^{n} a_{jr} \cos(\omega_r t - \delta_r)}$$

as amplitudes $(a_{1r}, \dots, a_{nr})$ están relacionadas entre si a través da ecuación característica

As frecuencias $\omega_r$ son solución do determinante secular.

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Para $n = 2$

$$q_1(t) = a_{11} \cos(\omega_1 t - \delta_1) + a_{12} \cos(\omega_2 t - \delta_2)$$

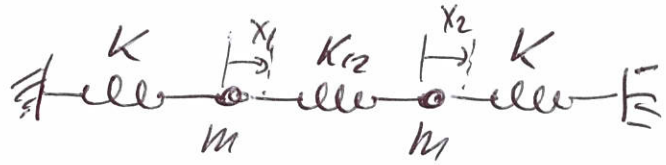
$$q_2(t) = a_{21} \cos(\omega_1 t - \delta_1) + a_{22} \cos(\omega_2 t - \delta_2)$$

$$\begin{vmatrix} A_{11} - \lambda^2 m_{11} & A_{12} - \lambda^2 m_{12} \\ A_{21} - \lambda^2 m_{21} & A_{22} - \lambda^2 m_{22} \end{vmatrix} = 0 \Rightarrow \begin{matrix} \lambda_1^2 \equiv \omega_1^2 \\ \lambda_2^2 \equiv \omega_2^2 \end{matrix}$$

$\omega_1$ $\quad \begin{pmatrix} & \\ & \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} = 0 \Rightarrow a_{21} = -\underbrace{f_1}_{\text{factor}} a_{11} \Rightarrow \vec{a}_1 \equiv \begin{pmatrix} a_{11} \\ -f_1 a_{11} \end{pmatrix}$

$\omega_2$ $\quad \begin{pmatrix} & \\ & \end{pmatrix} \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = 0 \Rightarrow a_{12} = -f_2 a_{22} \Rightarrow \vec{a}_2 \equiv \begin{pmatrix} -f_2 a_{22} \\ a_{22} \end{pmatrix}$

* No caso particular de $n=2$




$$T = \frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2$$

$$V = \frac{1}{2} (K + K_{12}) x_1^2 + \frac{1}{2} (K + K_{12}) x_2^2 - \frac{1}{2} 2 K_{12} x_1 x_2$$

$$m_{jk} = \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix} \qquad A_{jk} = \begin{pmatrix} K + K_{12} & -K_{12} \\ -K_{12} & K + K_{12} \end{pmatrix}$$

$$A - m\lambda^2 = \begin{pmatrix} K + K_{12} - m\lambda^2 & -K_{12} \\ -K_{12} & K + K_{12} - m\lambda^2 \end{pmatrix}$$

$$\det(A - m\lambda^2) = 0 \Rightarrow (K + K_{12} - m\lambda^2) = K_{12}^2 \Rightarrow \lambda^2 = \frac{K}{m} = \omega_1$$

$$\text{ou } \lambda_2^2 = \frac{K + 2K_{12}}{m} = \omega_2$$

$$\Rightarrow \omega_1 = \sqrt{\frac{K}{m}} \; ; \quad \omega_2 = \sqrt{\frac{K + 2K_{12}}{m}}$$

$\underline{\omega_1}$ $(A - m\omega_1^2)\vec{a}_1 = 0 \Rightarrow \begin{pmatrix} K_{12} & -K_{12} \\ -K_{12} & K_{12} \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} = 0 \Rightarrow a_{11} = a_{12}$

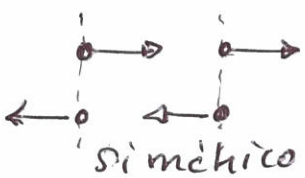
$\underline{\omega_2}$ $(A - m\omega_2^2)\vec{a}_2 = 0 \Rightarrow \begin{pmatrix} -K_{12} & -K_{12} \\ -K_{12} & -K_{12} \end{pmatrix} \begin{pmatrix} a_{21} \\ a_{22} \end{pmatrix} = 0 \Rightarrow a_{22} = -a_{21}$

$$\Rightarrow \quad x_1 = a_1 \cos(\omega_1 t + \varphi_1) + a_2 \cos(\omega_2 t + \varphi_2)$$

$$x_2 = a_1 \cos(\omega_1 t + \varphi_1) - a_2 \cos(\omega_2 t + \varphi_2)$$

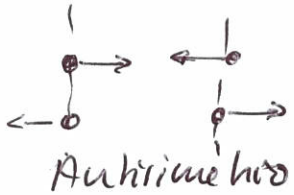
Modo $\omega_1$

$$\vec{a}_1 \begin{pmatrix} a_1 \\ a_1 \end{pmatrix}$$

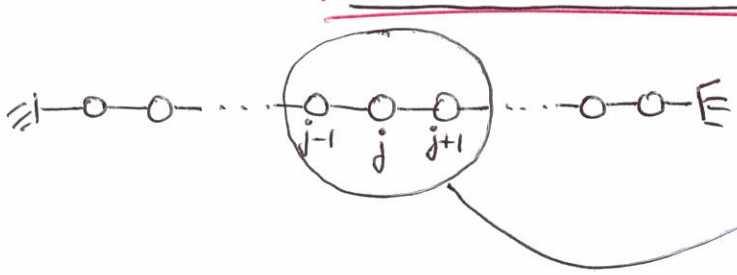


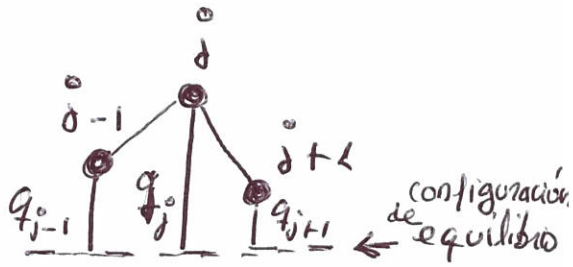

Modo $\omega_2$

$$\vec{a}_2 \begin{pmatrix} a_2 \\ -a_2 \end{pmatrix}$$



# Corda discreta




$$V = \sum_{j=1}^{n+1} \frac{1}{2} k \left( q_j - q_{j-1} \right)^2$$

$k = \hat{k} \frac{d-a}{d}$ ← longitude sen estirar; ← longitude en equilibrio.

$\hat{k}$ ← cte elástica ; $k = m\omega_0^2$

$$T = \sum_{j=1}^{n} \frac{1}{2} m \dot{q}_j^2$$ ← o sumatorio pódese extender a n+1 xa que $q_{n+1} = 0$.

Ecuación do movemento

$$m \ddot{q}_j = -m\omega_0^2 (q_j - q_{j-1}) + m\omega_0^2 (q_{j+1} - q_j) \quad j = 1, \dots, n$$

Forzas dos resortes veciños de j

Solución: $q_j = a_j e^{i\lambda t}$ ; os $a_j$ verifican:

$$(*) \quad -\omega_0^2 a_{j-1} + (-\lambda^2 + 2\omega_0^2) a_j - \omega_0^2 a_{j+1} = 0$$

A matriz $A - m\lambda^2$ tería a forma:

$$\begin{pmatrix} (\,) & -\omega_0^2 & 0 & 0 \\ -\omega_0^2 & (\,) & -\omega_0^2 & 0 \\ 0 & -\omega_0^2 & (\,) & -\omega_0^2 \\ 0 & 0 & -\omega_0^2 & (\,) \end{pmatrix} \vec{a} = 0$$

$-\lambda^2 + 2\omega_0^2$

$\rightarrow \exists$ Solución $\vec{a} \neq 0$ se $\det(A - m\lambda^2) = 0$

$\Rightarrow \lambda^2 \equiv \omega_r^2 \quad r = 1, \dots, n$

n frecuencias propias

A solución sería como no caso xeral:

$$q_i(t) = \sum_{r=1}^{n} a_{jr} \cos(\omega_r t - \delta_r)$$

Pero os $a_{jr}$ poden calcularse analíticamente seguindo o Maxwell.

$$a_{jr} = a_r e^{i(j\phi_r - \varepsilon_r)}$$

→ Substituindo na ec. característica (*) obtemos as frecuencias propias

$$\boxed{\omega_r = 2\omega_0 \operatorname{sen}\left(\frac{\phi_r}{2}\right)}$$

→ As condicións de contorno $a_{0r} = 0$ e $a_{n+1\,r} = 0$ determinan $\varepsilon_r = \frac{\pi}{2}$ e $\phi_r = \frac{r\pi}{n+1}$ respectivamente.

Tomando a parte real de $a_{jr}$:

$$a_{jr} = a_r \operatorname{sen}\left(j \frac{r\pi}{n+1}\right)$$

- $a_r$: condicións iniciais
- $j$: posición do oscilador
- $r$: modo normal
- $n$: nº de osciladores.

Solución final

$$\boxed{q_i(t) = \sum_{r=1}^{n} a_r \operatorname{sen}\left(j \frac{r\pi}{n+1}\right) \cos(\omega_r t - \delta_r)}$$

$$\text{sendo } \omega_r = 2\omega_0 \operatorname{sen}\left(\frac{r\pi}{2(n+1)}\right)$$

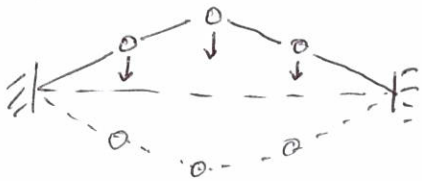
Exemplo: $\boxed{N = 3}$

$r = 1$

$q_1 = a_1 \operatorname{sen}\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen}\frac{\pi}{4} = \frac{\sqrt{2}}{2}$)

$q_2 = a_1 \operatorname{sen} 2\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 2\frac{\pi}{4} = 1$)

$q_3 = a_1 \operatorname{sen} 3\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 3\frac{\pi}{4} = \frac{\sqrt{2}}{2}$)



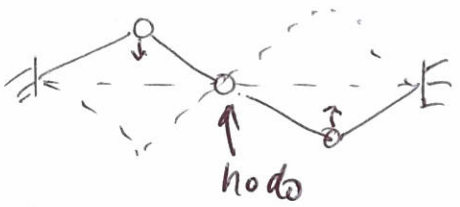
$$\vec{a}_1 \propto \begin{pmatrix} \frac{\sqrt{2}}{2} \\ 1 \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

$r = 2$

$q_1 = a_2 \operatorname{sen} 2\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 2\frac{\pi}{4} = 1$)

$q_2 = a_2 \operatorname{sen} 4\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 4\frac{\pi}{4} = 0$)

$q_3 = a_2 \operatorname{sen} 6\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 6\frac{\pi}{4} = -1$)



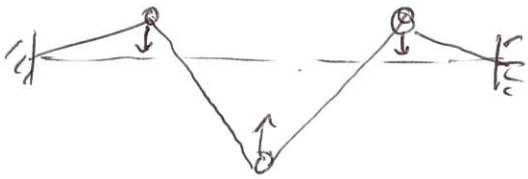

$$\vec{a}_2 \propto \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$r = 3$

$q_1 = a_3 \operatorname{sen} 3\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 3\frac{\pi}{4} = \frac{\sqrt{2}}{2}$)

$q_2 = a_3 \operatorname{sen} 6\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 6\frac{\pi}{4} = -1$)

$q_3 = a_3 \operatorname{sen} 9\frac{\pi}{4}\cos\omega_1 t$ ($\operatorname{sen} 9\frac{\pi}{4} = \frac{\sqrt{2}}{2}$)



$$\vec{a}_3 \propto \begin{pmatrix} \frac{\sqrt{2}}{2} \\ -1 \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$